

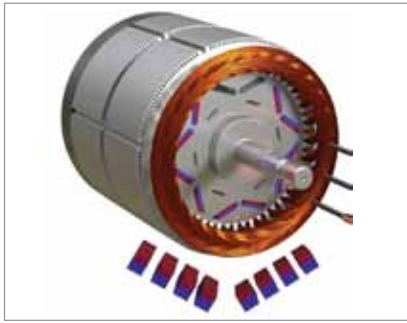


Fig. 7: An assistive permanent magnet synchronous reluctance motor

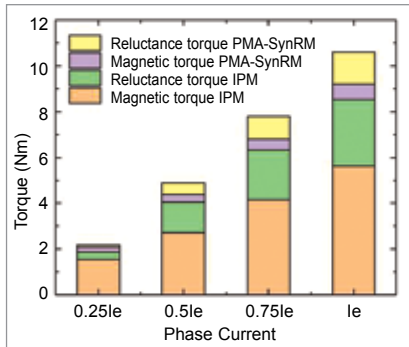


Fig. 8: Motor torque vs input current of a motor

- High-speed operation possible
- Their disadvantages:**
- Acoustically noisy
- High vibration
- Magnetic non-linearities make smooth torque control difficult
- Dependent on electronic control for operation

#### Their applications:

- Domestic appliances such as washing machines, vacuum cleaners, and fans

### IPM-SynRM electric motors

The Tesla internal permanent magnet synchronous reluctance motors (IPM-SynRM) are now used in some Tesla electric vehicles. Earlier, Tesla Motors used induction (or asynchronous) electric motors. From Model 3 onwards, the company has been using the IPM-SynRM motors, also known as a PMA-SynRM (permanent magnet assisted synchronous reluctance motors).

In general, it is a type that joins a motor type with an internal permanent magnet and a rotor type with a synchronous reluctance motor to fulfil the requirement characteristic in an electric vehicle (EV) application—high efficiency at high or low speeds. Tesla is not the first to utilise this design of

## Comparing Brushless Motors

Table 1

| Torque, speed, efficiency, and control       | IM      | SPM     | IPM    | SynRM   | SRM    |
|----------------------------------------------|---------|---------|--------|---------|--------|
| Peak efficiency                              | Average | Strong  | Strong | Good    | Good   |
| Efficiency flatness                          | Poor    | Average | Poor   | Good    | Good   |
| Constant power range                         | Poor    | Weak    | Poor   | Good    | Strong |
| Overload capability                          | Average | Good    | Good   | Average | Strong |
| Torque density                               | Average | Strong  | Strong | Good    | Good   |
| Power density                                | Poor    | Average | Good   | Good    | Good   |
| Stall resistance                             | Weak    | Strong  | Strong | Strong  | Strong |
| Torque ripple                                | Strong  | Poor    | Good   | Strong  | Weak   |
| Inverter VA required                         | Average | Good    | Good   | Weak    | Strong |
| Bus capacitor ripple                         | Strong  | Average | Good   | Strong  | Weak   |
| Sensorless operation                         | Poor    | Average | Good   | Good    | Good   |
| <b>Mechanical and environmental features</b> |         |         |        |         |        |
| Fault tolerance                              | Average | Weak    | Poor   | Average | Strong |
| High temperature use                         | Poor    | Weak    | Weak   | Good    | Strong |
| High-speed operation                         | Average | Weak    | Good   | Poor    | Strong |
| Harsh environment                            | Average | Weak    | Weak   | Good    | Strong |
| Small size and weight                        | Average | Strong  | Strong | Good    | Good   |
| Rare earth magnets                           | Strong  | Weak    | Weak   | Strong  | Strong |
| Rotor inertia                                | Poor    | Good    | Good   | Average | Good   |
| Ease of recycling                            | Good    | Poor    | Weak   | Strong  | Strong |
| Acoustic noise                               | Strong  | Average | Good   | Strong  | Weak   |
| Easy to repair                               | Poor    | Weak    | Weak   | Poor    | Strong |
| System cost                                  | Good    | Average | Poor   | Average | Good   |

motor, but its variant is thought of as one of the best simply due to the high efficiency and range of Tesla cars.

A specific innovation from Tesla is the use of segmented magnets of four parts instead of the more typical single fixed magnet inside the rotor of the motor. It helps reduce eddy currents and reduces the risk of magnets overheating.

Tesla Motor engineers made an impressive design choice when they developed the Tesla Model 3. They abandoned the conventionally used and proven induction motors and replaced them with a new type of motor, called the IPMSynRM motor. These motors have a totally different design, making use of magnetic action and reluctance. Tesla Motors began replacing the induction motors on their Model S and X with this new engine as well.

Each motor has a rotor and stator arrangement. In a simple induction motor, the rotating part of an engine is an arrangement of conducting bars. Since the stator creates a rotating

magnetic field, the floating field interacts with the rotor bars and generates currents in the rotor bars. The induced currents and the RMF interact, imposing force on the rotor bars, and the rotor begins to rotate.

But there is a problem for long drives at cruising speed; losing 3% to 4% of the energy to generate currents in the rotor bars is definitely not efficient. That is why, to overcome this problem, Tesla replaced this engine in RWD and AWD cars with IPM-SynRM engine for better efficiency. The IPM-SynRM is a combination of permanent magnet and SynRM motors, which are integrated by placing the permanent magnets in the grooved cuts of the SynRM motor, deep in the iron core.

Please see Table 1 for overall comparison of different types of motors. **EFY**

